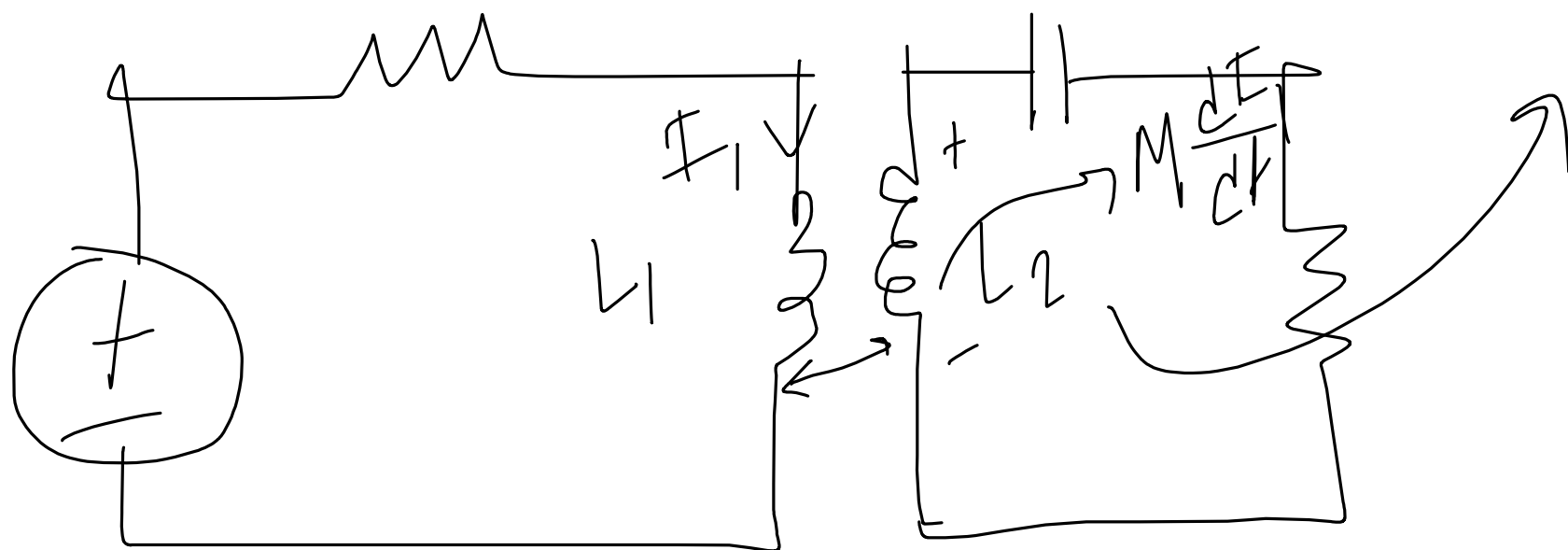
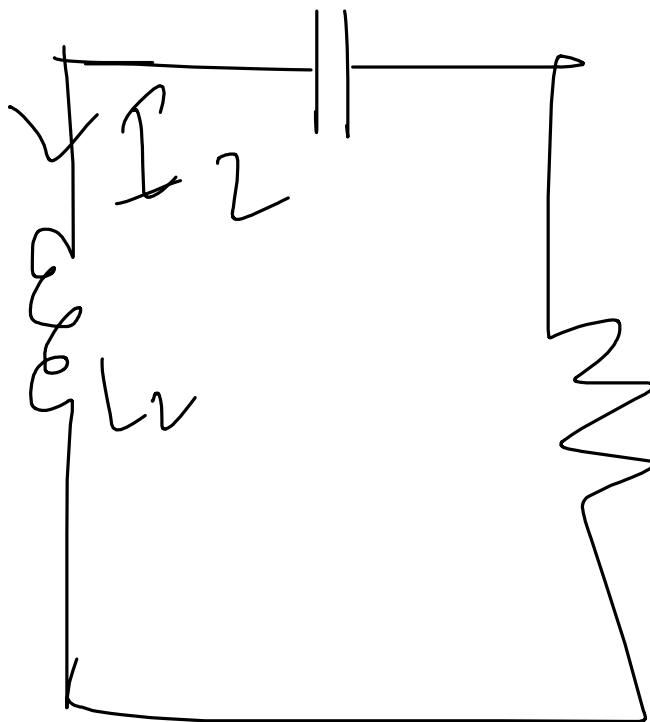
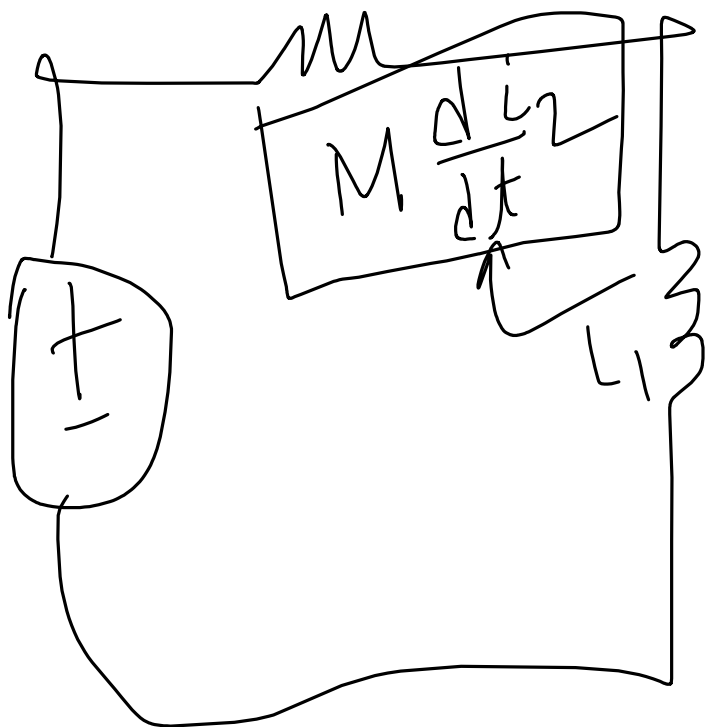


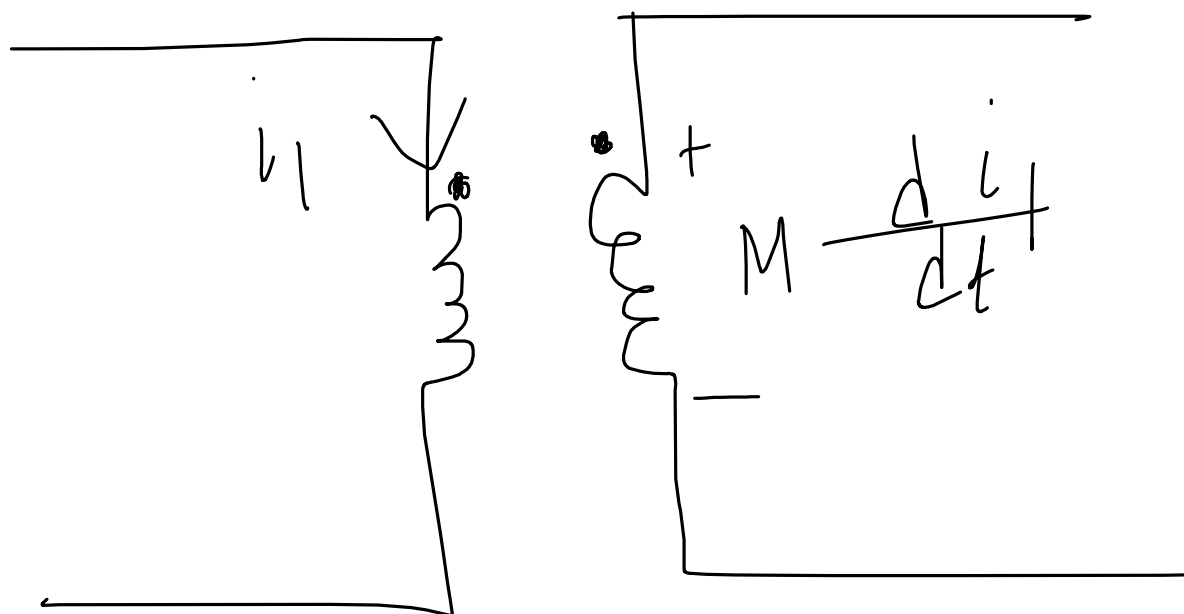
Mutual Coupling

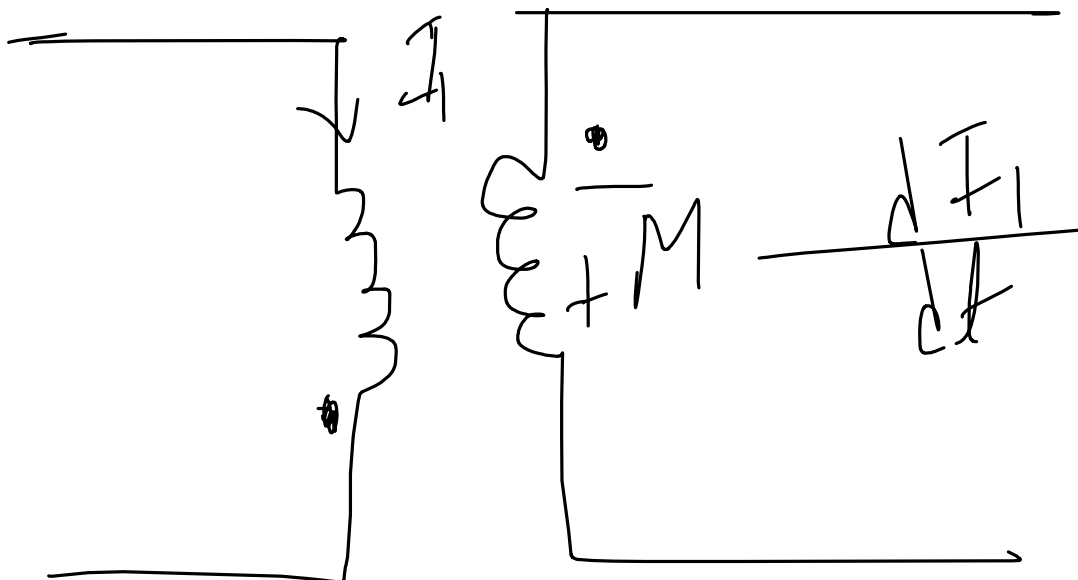


$\frac{+}{-}$ $\frac{+}{-}$



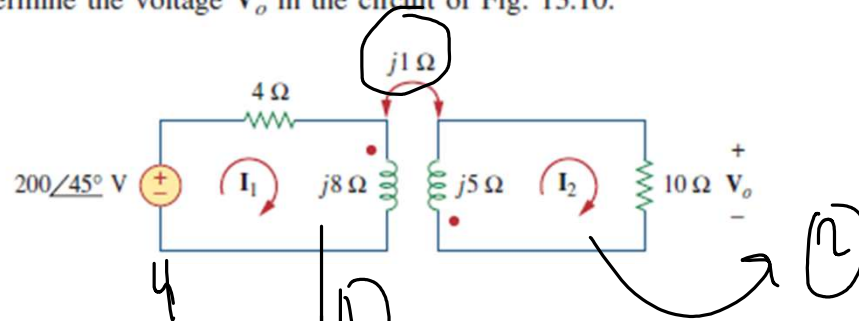
Dot convention



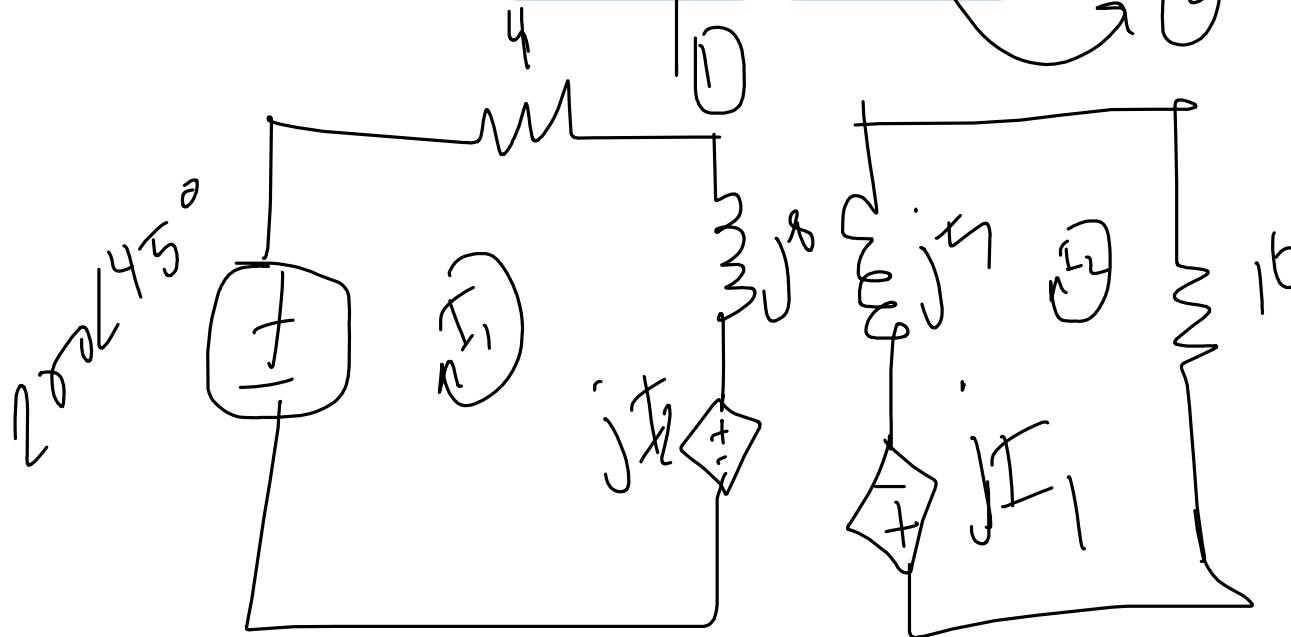


Practice Problem 13.1

Determine the voltage V_o in the circuit of Fig. 13.10.



$$V = I Z$$



KVL-1

$$-200 \angle 45^\circ + (4 + j8)I_1 + jI_2 = 0 \quad \text{--- (1)}$$

KVL-2

$$jI_1 + (10 + j5)I_2 = 0$$
$$I_2 = \frac{-jI_1}{10 + j5} \quad \text{--- (2)}$$



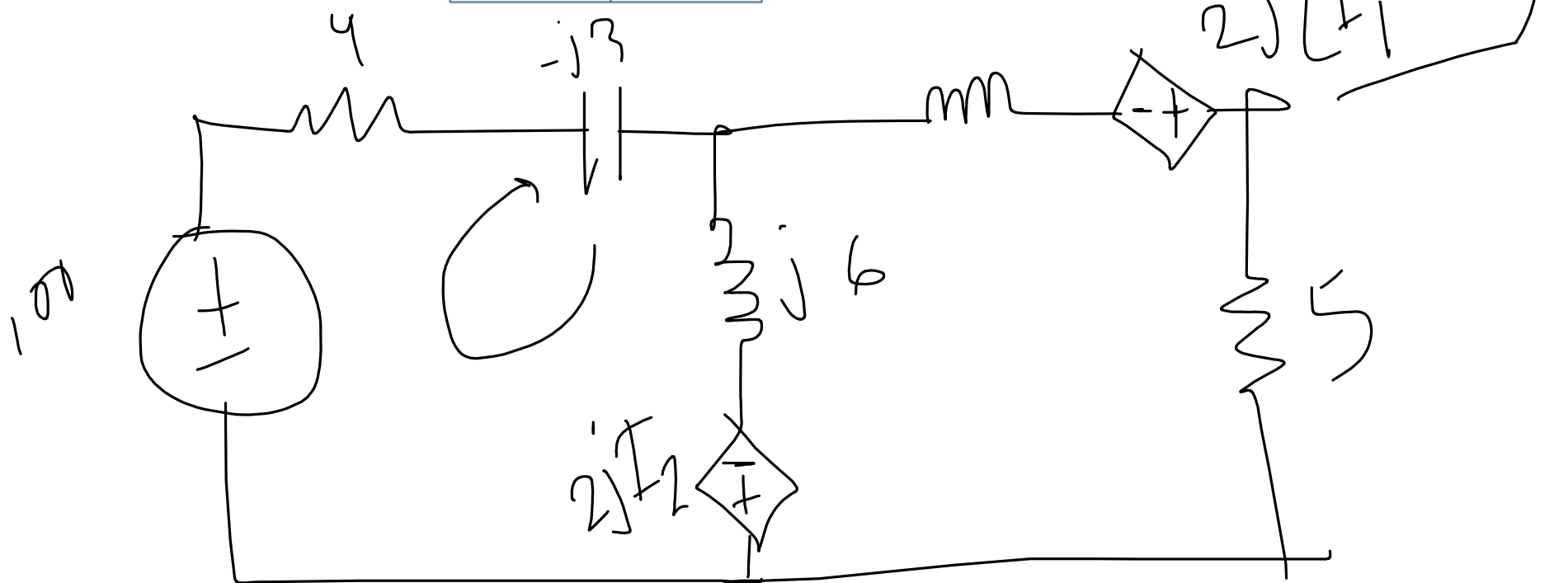
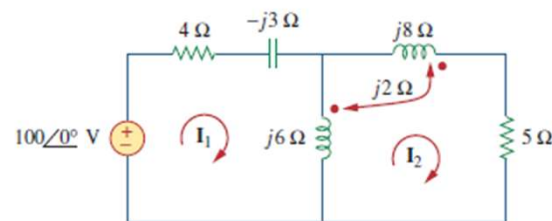
$$I_1 = 2L_{-134 \cdot 43^0}$$

$$\therefore I_2 = 22 \cdot 36L_{-17 \cdot 86^6}$$

$$V_0 = 2 \times 10^{-134.43}$$

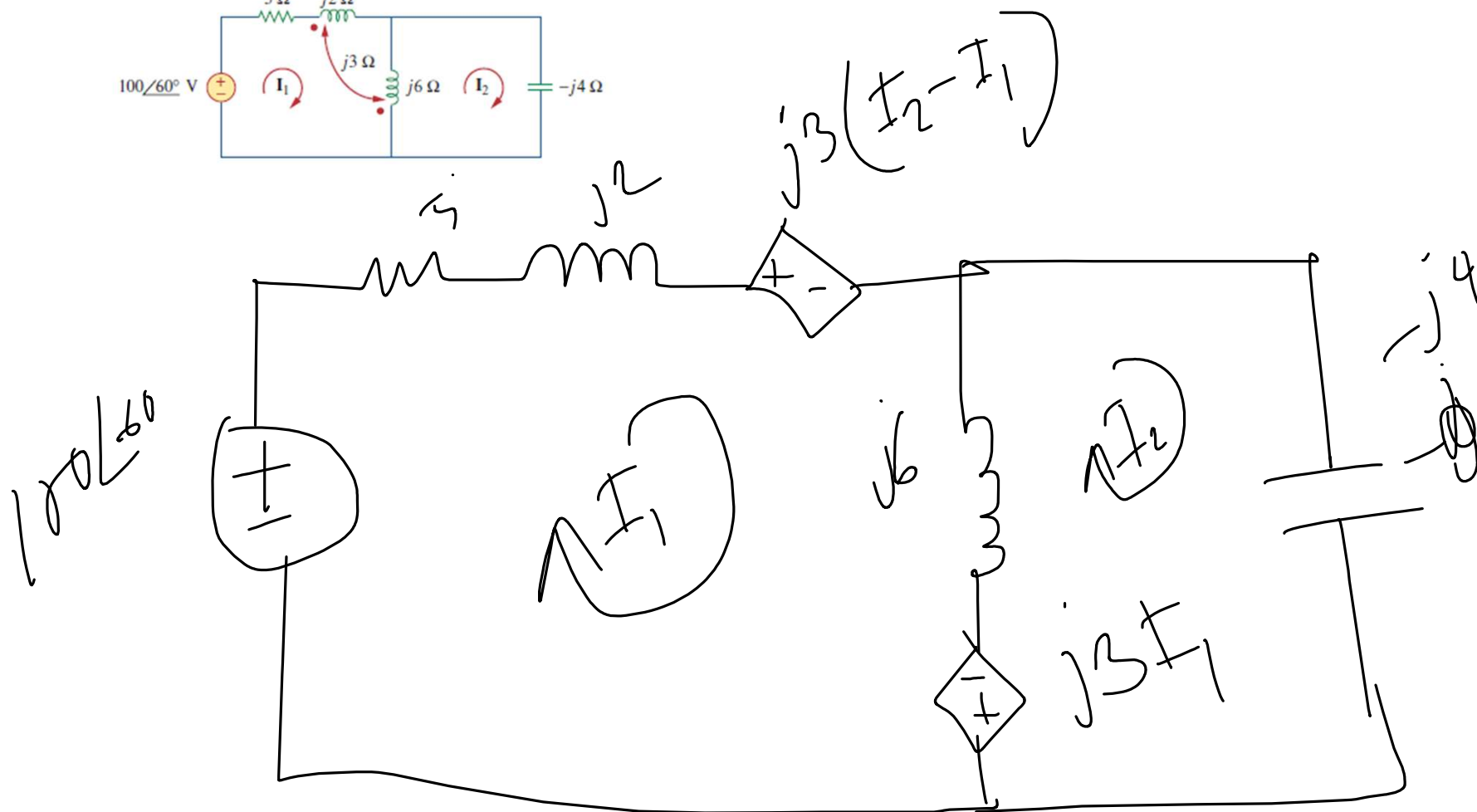
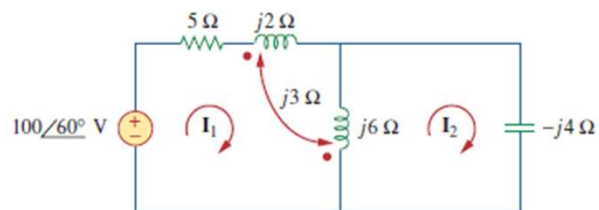
Example 13.2

Calculate the mesh currents in the circuit of Fig. 13.11.



Determine the phasor currents I_1 and I_2 in the circuit of Fig. 13.13.

Practice Problem 13.2



KVL-1

$$-100 \angle 60^\circ + (5 + j2)$$

$$) I_1 + j3(I_2 - I_1) = 0$$

$$+ j6(I_1 - I_2) - j3I_1 = 0$$

$$\Rightarrow I_1(5 + j2 - j3 + j6 - j3) + I_2(j3 - j6) = 0$$

$$\Rightarrow I_1(5 + j2) + I_2(-j3) = 0$$

$$\frac{KVL-2}{j3I_1 + j6(I_2 - I_1) + I_2(-j4) = 0}$$

$$j3I_1 + j6I_2 - j6I_1 - j4I_2 = 0$$

$$\rightarrow -I_1 \times j3 + I_2 \times 2j = 0$$

$$\therefore I_2 = \frac{j3I_1}{2j} = 1.5I_1$$

②

~~⇒ ①~~

$$I_1(5+2j)$$

$$+ I_1 \times 1.5(-j3)$$

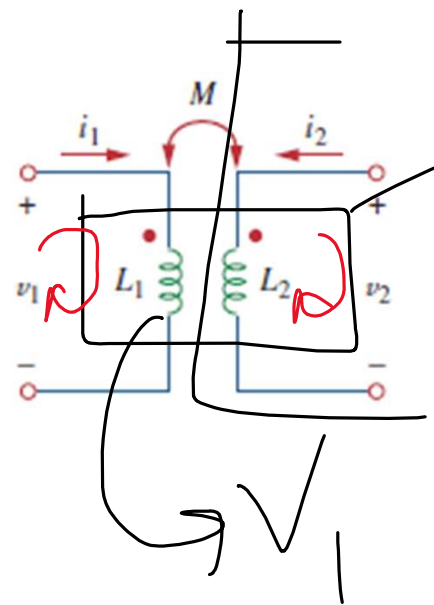
$$= 188 \angle 60^\circ$$

$$I_1 = 17.9 \angle 86.5^\circ$$

$$I_2 = 1.5 \times I_1$$

$$\frac{1}{2} L i^2$$

Energy in Coupled Ckt

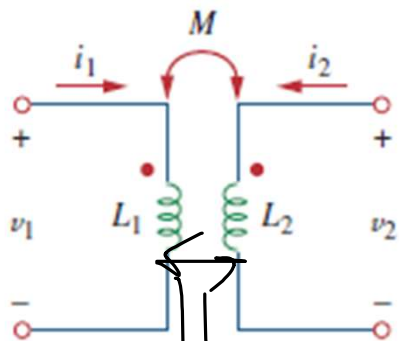


$$V_2$$

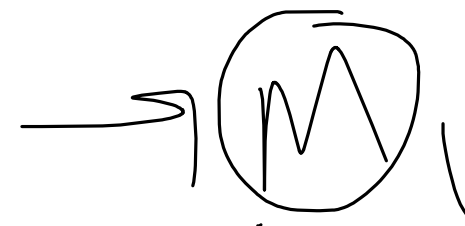
$$W = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M i_1 i_2$$

otherwise

if current in both coil enter or leave through dotted terminal



$$L = 1 \text{ cm}$$



$$L = 0.5 \text{ cm}$$



$$M = \sqrt{L_1 L_2}$$

coupling coefficient

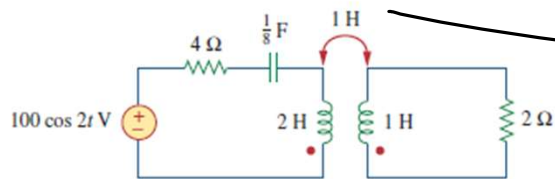
$0 \leq \sqrt{L_1 L_2} \leq 1$
unitless

A handwritten diagram illustrating the relationship between the mutual inductance M , the coupling coefficient, and a unitless expression. The equation $M = \sqrt{L_1 L_2}$ is written at the top. A large curved arrow originates from the M in this equation and points to a smaller version of the same equation, $0 \leq \sqrt{L_1 L_2} \leq 1$, which is labeled "unitless". A straight arrow points from the $\sqrt{L_1 L_2}$ term in the top equation to the text "coupling coefficient".

$$K = \frac{\cancel{Q_2} \cancel{W}}{\cancel{Q_1} + \cancel{Q_2}}$$

For the circuit in Fig. 13.18, determine the coupling coefficient and the energy stored in the coupled inductors at $t = 1.5$ s.

Practice Problem 13.3



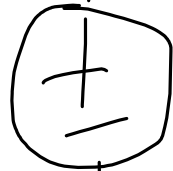
$$w = 2$$

$$k = 1$$

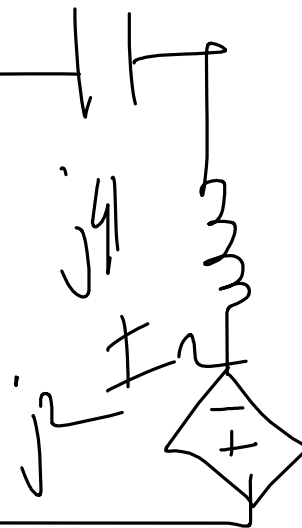
$$W$$

$$j\omega L_1$$

$$100 \angle 0^\circ$$



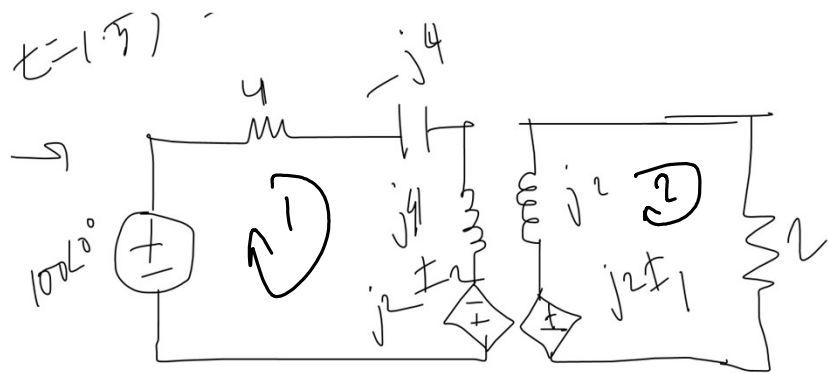
$$j\omega L_2$$



$$j\omega L_1$$

$$j\omega L_2$$

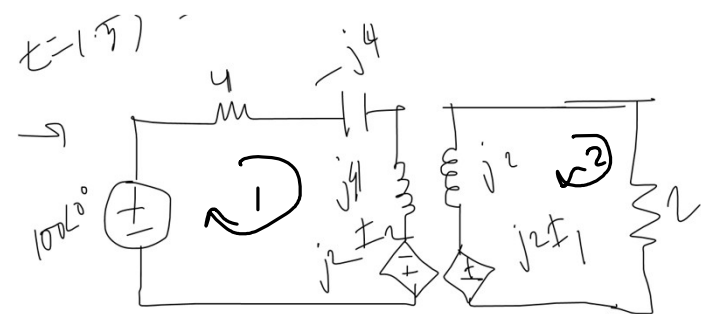
$$2$$



KVL-1

$$-100\angle 0^\circ + [4 - j4 + j4] I_1 - I_2 \times j2 = 0$$

$$\Rightarrow 4I_1 - j2I_2 = 100\angle 0^\circ \quad \text{--- (1)}$$



$$\text{KVL-2}$$

$$-2jI_1 + (2 + j2)I_2 = 0$$

$$\Rightarrow I_2 = \frac{2jI_1}{2 + j2}$$

$$\Rightarrow I_2 = \frac{0.5 + 0.5j}{1} I_1$$

② $\Rightarrow$ ①

$$4I_1 - j2 \times 10^{-3} + j0.5 I_1 = 100 \angle 0^\circ$$

$$4I_1 - (1j - 1) I_1 = 100 \angle 0^\circ$$

$$I_1 = \frac{100 \angle 0^\circ}{5 - j} = 19.61 \angle 11.31^\circ$$

$$I_2 = 13.87 \angle 56.3^\circ$$

$$w = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 \pm M i_1 i_2$$

$$i_1 = 19.6 \angle \cos(2t + 11.3^\circ)$$

$$i_2 = 13.87 \angle \cos(2t + 56.3^\circ)$$

$2t = 2x$

$$2t = 2 \times 1.5$$

$$= 3 \text{ rad}$$

$$V = 1000 \cos(2t)$$

$$\pi = 180^\circ$$

$$3 \text{ rad} =$$

$$\frac{180}{\pi} \times 3$$

$$= 171.89^\circ$$

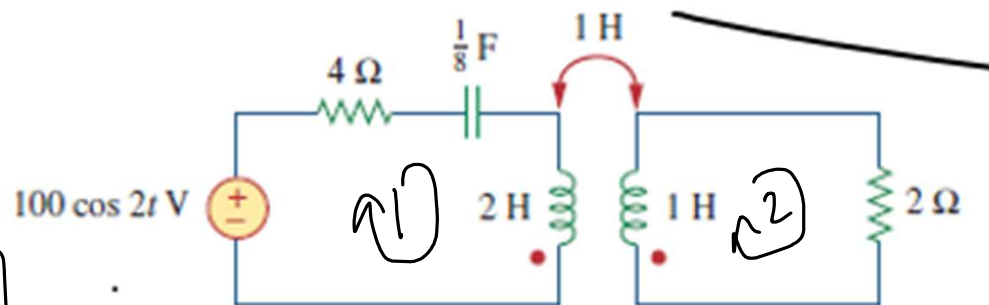
rad/s

$$t = 1.7 \text{ ns}$$

$$\hookrightarrow i_1 = 19.58 \text{ A}$$

$$i_2 = 9.25 \text{ A}$$

$$w = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 \pm M i_1 i_2$$



Coil-1 → leave
Coil-2 → enter

$$\therefore W = \frac{1}{2} \times 2 \times 19.38^2 + \frac{1}{2} \times 1 \times 9.25^2 - 1 \times 19.38 \times 9.25$$

$$= 245.03 \text{ J} \quad \checkmark$$

$$M = \sqrt{L_1 L_2}$$

$$\frac{1}{M} = \frac{1}{\sqrt{L_1 L_2}}$$

$$= \frac{1}{\sqrt{2 \times 1}}$$